A Novel High-Speed 4-bit Carry Generator with a New Structure for Arithmetic Operations

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***Abstract*—A new high speed 4-bit carry generator is proposed in this paper. A new structure has been introduced to reduce the delay from inputs to the outputs. Also this structure utilizes less number of transistors and it occupies smaller area size. Proposed structure has been simulated by HSPICE software in a typical 0.18 µm CMOS technology and results show that there is a 500ps delay from inputs to output.**

***Keywords*—adder;carry delay; carry look ahead; high speed; multiplier**

I. INTRODUCTION

Fast adders are one of the widely used elements in computer systems. Various adder structures are used to implement addition [1-2]. The simplest and the easiest binary adder to be implemented is ripple carry adder [3-5]. A ripple carry adder is a circuit that produces the sum of two binary numbers. This adder can be constructed with cascaded full adders. In this structure, the carry output from each full adder goes to the carry input of the next full adder. The ripple carry has a long circuit delay due to the carry propagation delay and this delay deteriorates with the increase in the number of input bits.

Carry Select Adders (CSA) are adders which are faster compared to ripple carry adders [6]. In these kinds of adders, two Ripple Carry Adders (RCA) are used. The first RCA has the input carry equal to zero and the second one have the input carry equal to 1. A multiplexer (MUX) is used to select the correct result based on the carry of the previous digit. Fig. 1shows a block diagram of a CSA:

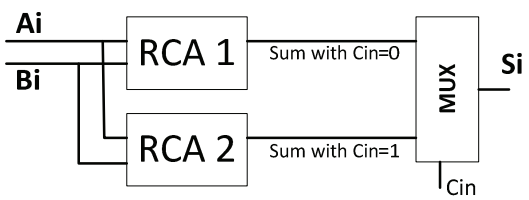


Fig. 1. Block diagram of a Carry Select Adder

Another structure of adders which are faster than ripple carry adders are carry look ahead adders[7-11]. These adders are widely used due to their superior performance over ripple carry adders. Using the Propagate (P_i) and Generate (G_i) logic to make addition much faster, the Carry Look Ahead Adder (CLA) is able to generate carries before the sum is produced. Carry Generate and carry Propagate functions for A_i and B_i as inputs at bit position *i* can be calculated from equation (1) and (2).

$$G_i = A_i.B_i \quad (1)$$

$$P_i = A_i \oplus B_i \quad (2)$$

Using C_i as the carry input, the sum and output carry can be written as follows:

$$C_{i+1} = G_i + P_i C_i \quad (3)$$

$$S_i = P_i \oplus C_i \quad (4)$$

Output carry for a 4-bit adder can be calculated as:

$$C_4 = G_3+P_3.G_2 + P_3.P_2.G_1 + P_3P_2.P_1.G_0 + P_3P_2.P_1.P_0.C_0 \quad (5)$$

The most important factor which must be considered in designing the adders is the speed of execution. The speed of the carry-propagation determines the speed of an adder to a large extend. The new structure proposed in this paper is used to design a high speed carry generator to be used in high speed adders. This paper is organized as follows: in section 2 proposed carry generator structures is presented; in section 3 simulation results are given. Finally, section 4 concludes the paper.

II. PROPOSED CIRCUIT

The proposed carry generator is designed based on the circuit shown in Fig. 2. Truth table of the circuit shown in Fig. 2 is given in Table I.

According to Table I, if the number of inputs with binary value of 1 is more than or equal to 2, then *Carry* = 1 otherwise it will be 0. Let *N* be the number of inputs with binary value of 1. In the circuit shown in Fig. 2, if $N \geq 2$, then the number of ON NMOS transistors will be more than ON PMOS transistors. Hence the voltage of node D will become less than $V_{dd}/2$ and *Carry* will have logic value of 1. On the other hand, if $N \leq 1$, then the number of ON PMOS transistors will be more than ON NMOS transistors and the voltage of node D will become more than $V_{dd}/2$. Consequently, *Carry* will have logic value of 0.

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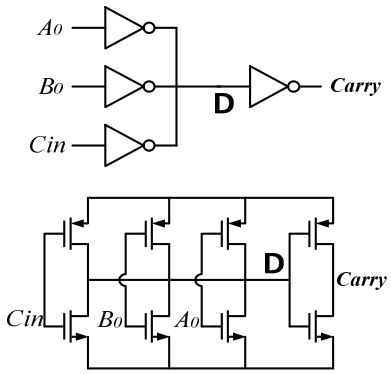


Fig. 2. 1-bit carry generator

TABLE I.
THE TRUTH TABLE OF CIRCUIT SHOWN IN FIG. 2

A	*B*	*Cin*	*Carry*
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

A new 4-bit carry generator is designed as shown in Fig. 3. Transistors are binary scaled and x is the ratio of μ_n (mobility of electrons) to μ_p (mobility of holes). The truth Table is given in Table II.

TABLE II.
THE TRUTH TABLE OF THE PROPOSED ADDER

Cin	*A0 B0*		*A1 B1*		*A2 B2*		*A3 B3*		*Carry*
x	x	x	x	x	x	x	1	1	1
x	x	x	x	x	x	x	0	0	0
x	x	x	x	x	1	1	$A_3 \neq B_3$		1
x	x	x	x	x	0	0	$A_3 \neq B_3$		0
x	x	x	1	1	$A_2 \neq B_2$		$A_3 \neq B_3$		1
x	x	x	0	0	$A_2 \neq B_2$		$A_3 \neq B_3$		0
x	1	1	$A_1 \neq B_1$		$A_2 \neq B_2$		$A_3 \neq B_3$		1
x	0	0	$A_1 \neq B_1$		$A_2 \neq B_2$		$A_3 \neq B_3$		0
1	$A_0 \neq B_0$		$A_1 \neq B_1$		$A_2 \neq B_2$		$A_3 \neq B_3$		1
0	$A_0 \neq B_0$		$A_1 \neq B_1$		$A_2 \neq B_2$		$A_3 \neq B_3$		0

According to the truth table if $A_3 = B_3$ and both of them have binary value equal to 1, then *Carry* = 1. In the proposed circuit shown in Fig. 3 if $A_3 = B_3 = 1$, then branch 3 of the NMOS will be turned ON and branch 3 of PMOS transistors will be turned OFF. Since transistor size of the branch 3 of NMOS transistors is more than the sum of the PMOS transistors in branch 0,1 and 2, hence the voltage of node D will become less than $V_{dd}/2$ and *Carry* will have binary value of 1. If $A_3 = B_3 = 0$, then branch 3 of PMOS transistors will be turned on. In this case the voltage of node D will become more than $V_{dd}/2$ and the output will have the binary value equal to 0.

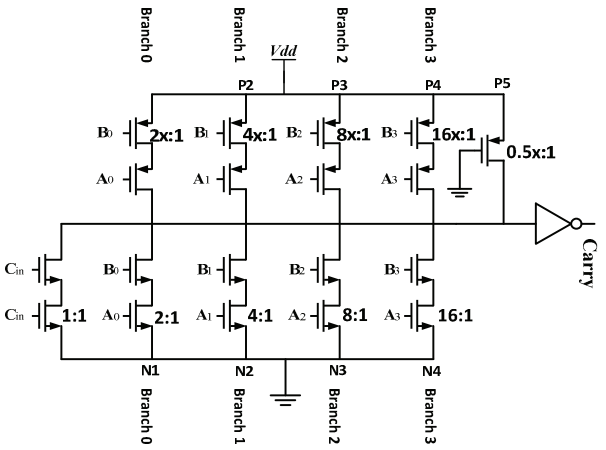


Fig. 3. Proposed 4-bit carry generator

For the other cases, for example when $A_3 \neq B_3$, $A_2 \neq B_2$ and $A_1 = B_1 = 1$, *Carry* will have binary value of 1. In this case branch 1 of NMOS transistors will become ON and branch 1,2 and 3 of PMOS transistors and also branch 2 and 3 of NMOS transistors wil become OFF. Thus the voltage of node D will become less than $V_{dd}/2$ and the output will have the binary value equal to 1. The function of the circuit in other cases is like the function of the circuit in the given example. There is one special case and it happens when $A_3 \neq B_3$, $A_2 \neq B_2$, $A_1 \neq B_2$, $A_0 \neq B_0$ and $C_{in} = 0$. In this case the output is supposed to have binary value of 0. All branches of PMOS and NMOS transistors are OFF and node D is in the float mode. For this reason a minimum size transistor P_5 is used to increase the voltage of node D to a value more than $V_{dd}/2$ which will lead to an output with a binary value of 0. This case is the worst case with the lowest speed for the designed circuit. By applying some minor changes in the circuit and doubling the size of transistors, we can increase the speed of the circuit.

The behavior of the transistor P_5 is as follows:In the circuit shown in Fig. 3 if $C_{in} = 0$ and $A_{0\text{-}3} \neq B_{0\text{-}3}$, then P_5 should increase the voltage of node D to a value more than $V_{dd}/2$ and in other cases P_5 should have no effect on the circuit. Therefore, gate control voltage of P_5 is implemented through $A_0 \oplus B_0$ andC_{in}=0 (the added circuit is substituted branch 0 of NMOS transistors and transistor P_5) functions and this transistor is assumed to be turned on just in this case. Hence, size of the P_5 can be increased two times and it can be turned off in other cases which will result in a higher speed and lower power consumption. Improved circuit of the Fig. 3 is shown in Fig. 4.

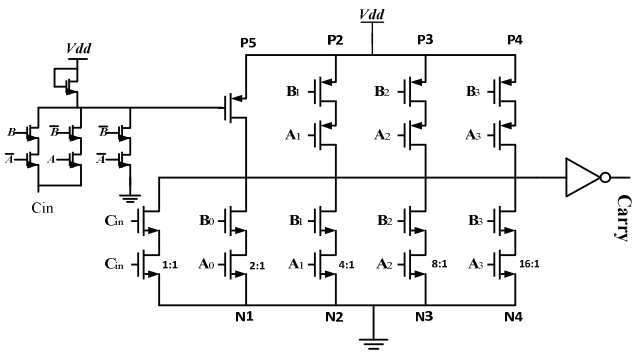


Fig. 4. Improved 4-bit carry generator

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As can be seen, the proposed carry generator circuit utilizes less number of transistors compared to other carry look ahead topologies. Also, input of the circuit is directly from the input data and there is no need to produce Generate and Propagate functions. This results in a lower latency from inputs to outputs and also lower size of the circuit and power consumption.

III. Simulation Results

Carry generator to be used in adders has been designed in a typical 0.18 μm CMOS process and simulated by HSPICE software.

Simulation results for the circuit shown in Fig. 4 is given in Fig. 5. The last row in Fig. 5 shows output carry for a 4-bit carry generator with Cin=0.

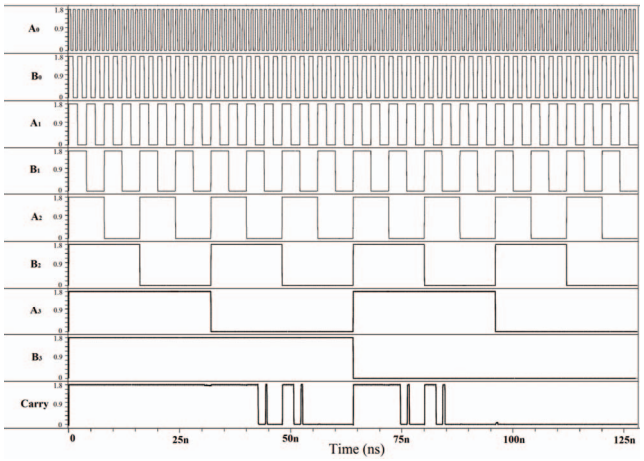


Fig. 5. Output carry for a 4-bit carry generator with C_{in}=0

Fig. 6 shows the simulation results for the proposed circuit in the worst case after the improvement of the circuit shown in Fig. 3. As can be seen from Fig. 6 there is a 256ps delay from inputs to output.

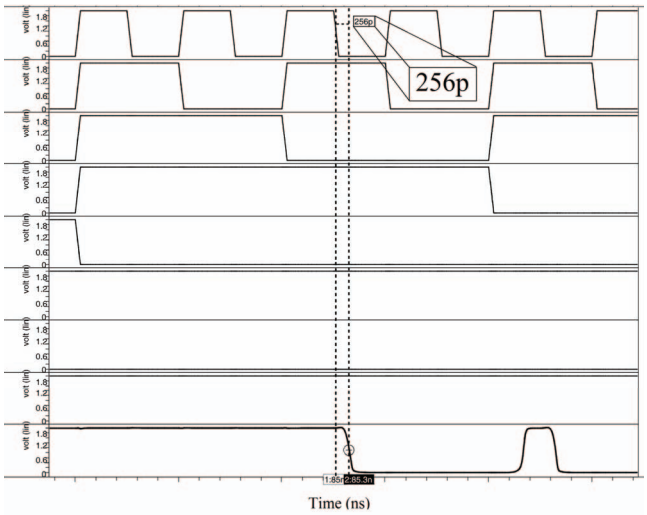


Fig. 6. Delay in the worst case for the improved 4-bit carry generator shown in fig. 4

IV. Conclusion

A new high-speed 4-bit carry generator using a new structure is proposed to be used in high-speed arithmetic operations and also in high-speed multipliers. The main advantage of the proposed structure is the low delay from inputs to outputs. In addition to the mentioned benefit, the designed carry generator utilizes less number of transistors which will result in lower power consumption.

Simulation results by HSPICE indicate that the delay from inputs to outputs is less than500ps for a typical 0.18μm CMOS technology at the supply voltage of 1.8V.

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